# HarbourOS: Enterprise Multi-Tenant Fisheries SaaS Strategy, Software Requirements Specification (SRS), and Microservices Build Plan

This deliverable provides the complete commercial, regulatory, architectural, and engineering blueprint for **HarbourOS**—a multi-tenant, cloud-edge SaaS platform and embeddable SDK ecosystem for fishing harbours, landing centres, cooperatives, and seafood exchanges[1]. Engineered for immediate operational and statutory fit in Kerala, India, the platform is architected to scale horizontally across Indian maritime states and international seafood markets[1].

To satisfy your request for two comprehensive Markdown documents within a single cohesive output, this report is organized into two standalone sections:

1. **FILE 1: HARBOUROS_FULL_SAAS_PLAN.md** — Details the Kerala-to-Global market strategy, statutory compliance under *The Kerala Fish Procurement, Marketing and Maintenance of Quality Act, 2021* (Act No. 16 of 2021)[4], the variable multi-party monetization and split-settlement model (covering both HarbourOS-hosted auctions and external harbour websites connected via our SDK/API)[6], frictionless onboarding, role-based mobile-first UX/UI plans, and the Super Admin control plane[1].
2. **FILE 2: HARBOUROS_SRS_AND_BUILD_PLAN.md** — Delivers the formal Software Requirements Specification (SRS), the pre-refactored Nx Monorepo microservices architecture, database connection pooling configurations, complete PostgreSQL 16 DDL schemas with Row-Level Security (RLS) and SHA-256 hash-chained double-entry accounting triggers[7], external SDK and Razorpay Route integration code, a phased execution roadmap, and the copy-pasteable **Master AI Engineering Prompt**.

# FILE 1: HARBOUROS_FULL_SAAS_PLAN.md

## 1. Executive Vision & The "Kerala-First, Global-Ready" Thesis

Digitizing primary seafood trade requires engineering for a high-velocity, low-trust physical workflow that unfolds between 4:30 AM and 8:00 AM in wet, corrosive, and intermittently connected coastal environments[1]. According to the *Kerala Development Report 2026* (reporting 2024–25 state figures), Kerala's marine fisheries sector encompasses **335 fishing villages**, **688 fishing landing centres**, **7 major fishing harbours**, and **14 minor fishing harbours** (21 harbours total), supporting approximately **8.24 lakh marine fisherfolk** and

producing **6.5 lakh metric tonnes (MT)** of marine fish annually[1]. Despite the scale of this coastal economy, a 2024 National Council of Applied Economic Research (NCAER) publication (analyzing 2022 survey data) found that only **12%** of surveyed fishermen had selling activities connected to the web-based Fish Market Price Information System (FMPIS)[1].

A critical strategic insight is that HarbourOS must not waste capital duplicating government identity or vessel registries[1]. The Government of India's **National Fisheries Digital Platform (NFDP)**—launched in September 2024 under the PM-MKSSY sub-scheme and updated through August 2026—already records **39,46,756 individual registrations** (25,91,139 male, 13,55,613 female, and 4 transgender) and **36,986 organizational registrations** (22,660 enterprises, 11,620 cooperatives, and 2,706 FFPOs), alongside 12 onboarded nationalized banks for credit facilitation[11]. Similarly, **ReALCraft** handles vessel registration and licensing, **SELF** manages fuel permits in Kerala, and Kerala's Harbour Engineering Department published an Expression of Interest (EOI) in April 2026 (with a tender value of ₹12.5 lakh) for a fishing harbour digital asset management portal[1]. Furthermore, early state-level digital fish trade initiatives such as **FERI** (Online Fish Trade and Online Fish Sales, reported in September 2022) and the national **ONDC** network operate in adjacent commerce spaces[1].

None of these public registries or consumer marketplaces solve the sub-second dockside clearing problem: **Bluetooth-linked catch weighing, crate-level QR lot serialization, hybrid floor-shout and remote clock bidding, real-time buyer credit-limit enforcement, and tamper-proof multi-party financial settlement**[1]. HarbourOS positions itself as the **operational and financial clearinghouse** connecting fishermen, licensed auctioneers, wholesale buyers, weighing clerks, and harbour management societies[1]. By uniting the back-office ERP discipline of Belgium's **Aucxis NEPTUNE**[3], the virtual clock-auction network model of Europe's **PEFA**[2], and the rapid physical inspection and hybrid remote bidding workflows of Tokyo's **Toyosu Market** and South Korea's **Busan Cooperative Fish Market**[10], HarbourOS delivers a localizedKerala solution that can expand nationwide and globally.

EXECUTIVE MARKET & ARCHITECTURE SYNTHESIS

# HarbourOS Captures the Operational Clearing Gap Across 709 Kerala Landing Nodes

## 01 PHYSICAL INFRASTRUCTURE VS. DIGITAL BASELINE

Kerala Marine Sector Scale (688 Landing Centres + 21 Harbours = 709 Nodes)

| LANDING CENTRES | FISHING HARBOURS | MARINE FISHERFOLK | FMPIS WEB ADOPTION |
|---|---|---|---|
| 688 | 21 | 8.24 Lakh | 12% |
| Scalable edge nodes | 7 Major · 14 Minor | 6.5 Lakh MT annual catch | 2022 survey baseline (NCAER) |

## 02 ARCHITECTURAL CAPABILITY & WHITESPACE MATRIX

Native / Automated · Partial / High · Context / Limited · None (Gap)

| SYSTEM CAPABILITY | PUBLIC REGISTRIES (NFDP / REALCRAFT) | LOCAL APPS (FERI) | GLOBAL INCUMBENTS (PEFA / AUCXIS) | HARBOUROS ARCHITECTURE |
|---|---|---|---|---|
| Fisher & Vessel Identity Registry | Core (NFDP/ReALCraft) | Partial | Ref Only (PEFA eLog) | API Sync / Reference |
| Offline-First Weighing & QR Lot Serialization | None | Limited | High (Aucxis NEPTUNE) | Native (CRDT + BLE) |
| Hybrid Shout + Remote Dutch Clock Bidding | Stated Goal | Basic Web | High (PEFA / KOSMOS) | Sub-50ms WebSocket + Floor App |
| Kerala Act 16 of 2021 4-Way Commission Split | None | Unverified | EU/JP Rules Only | Automated Paise-Exact Engine |
| Hash-Chained Double-Entry Ledger & Escrow | None | None | Bank Guarantee Manual | SHA-256 Immutable + Razorpay Route |
| External Harbour Website SDK & Webhook API | None | None | Proprietary Terminal | Plug-and-Play JS/Mobile SDK |

**Statutory & Clearing Whitespace:** Public platforms (NFDP/ReALCraft) handle identity registries, while global clocks (PEFA/Aucxis) lack Kerala Act 16 of 2021 4-way split rules and Razorpay Route escrow.

Hover/click any row to inspect gap

## 03 ANNUAL RECURRING REVENUE (ARR): PURE SAAS VS. HYBRID CLEARING CUT

Pure SaaS Subscription (₹10k/mo/site) · Hybrid SaaS + Clearing Cut




**Selected: 30 Sites (Regional Expansion)**

₹36.0L Fixed SaaS + ₹216.0L Clearing Cut = ₹252.0L ARR (7.0x Pure SaaS)

Market baseline figures reflect 2024–25 state data from the Kerala Development Report 2026 and NCAER survey findings. Capability comparisons contrast public infrastructure and legacy auction tools against the proposed HarbourOS architecture.

**Data sources:** National Fisheries Digital Platform (NFDP) · PEFA Auction Systems · Aucxis NEPTUNE & KOSMOS · Kerala Development Report 2026 & NCAER Survey

# 2. Regulatory Architecture & The "Win-Win" Variable Monetization Engine

## 2.1 Statutory Guardrails Under Kerala Act No. 16 of 2021

In Kerala, any software that touches fish auctioning or settlement must strictly comply with **The Kerala Fish Procurement, Marketing and Maintenance of Quality Act, 2021 (Act No. 16 of 2021)**[4]. Designing an arbitrary "platform commission" deducted from the fisherman's sale price would violate state law[1]. Specifically, the Act establishes six foundational legal constraints that HarbourOS encodes directly into its domain logic:

- **Notified Locations & Society Governance (Sections 3, 4, 7, 9 & 11)**: Fish auctioning is permitted only at government-notified fish landing centres, fishing harbours, and fish markets overseen respectively by a **Fish Landing Centre Management Society (FLCMS)**, **Harbour Management Society (HMS)**, or **Fish Market Management Society (FMMS)**, without restricting fishermen's right to sell fish directly without an auction[5]. Under Section 10, in special circumstances and with Government permission, the HMS may take over the first sale of fish by avoiding auction[17].
- **Auctioneer Licensing & Location Binding (Section 5)**: No person may conduct an auction without a valid 3-year licence issued by the Authorised Officer (an officer in the rank of Assistant Director of Fisheries) within 30 days of application, backed by a security deposit and a recommendation from the relevant HMS, FLCMS, FMMS, or local **Fisherman Development Welfare Co-operative Society**[5]. Each licence is legally bound to a specific district and named harbour, landing centre, or market[5].
- **Tied Co-operative Society Loan Rule (Section 4(8))**: Any fisherman who has taken a loan from a Fishermen Development Welfare Co-operative Society to purchase fishing equipment is legally required to auction their catch solely through an auctioneer deputed by that society[5].
- **Strict 5% Auction Commission Ceiling & Prohibition of Extra Benefits (Section 4(4) & 4(6))**: Total auction commission ("Tharaku", defined as the service charge realized from fish sellers for conducting or providing facilities for the auction) cannot exceed **five percent (5% / 500 basis points)** of the gross auction amount[5]. Section 4(6) strictly forbids the auctioneer or any other person from realizing any additional monetary or non-monetary benefit in connection with fish auctioning[5].
- **Mandatory Statutory Apportionment of the 5% Commission (Section 4(5))**:
  - **At Fishing Harbours and Fish Landing Centres**: The realized commission must be apportioned **20%** to the Auctioneer (1.0% of gross sale), **20%** to the HMS or FLCMS (1.0% of gross sale), **40%** returned to the fishermen concerned as an **Annual Production Bonus** (2.0% of gross sale), and **20%** to the co-operative societies participating in the auction (1.0% of gross sale)[5].
  - **At Public / LSGI Fish Markets**: The realized commission must be apportioned **20%** to the Auctioneer, **20%** to the FMMS, **30%** to the Fishermen Welfare Fund Board for welfare activities, and **30%** to the Local Self-Government Institution (LSGI) or market

owner[5].

- **Basic Floor Price & Quality Certification (Sections 16, 19 & 21)**: Under Section 19, the HMS holds statutory power to fix the basic (floor) price of fish from time to time, and fishermen may opt to have their catch procured through their Fishermen Development Welfare Co-operative Society at that basic price[5]. Under Section 21, quality certificates are issued by an authorized officer not below the rank of Fisheries Extension Officer based on standards set by the State Level Fish Quality Maintenance Committee (constituted under Section 16)[5].

## 2.2 How HarbourOS Takes Its Variable Cut While Keeping All Parties Happy

To ensure HarbourOS gets paid reliably across both **HarbourOS-hosted auctions** and **external harbour websites using our SDK/APIs**, while keeping fishermen, auctioneers, societies, and buyers happy and maintaining compliance with Section 4(4)–(6) of Act 16 of 2021, the platform uses a **Three-Stream Variable Revenue Architecture** (subject to local legal sign-off prior to deployment)[1]:

1. **Fishermen & Boat Owners Get 100% Statutory Protection**: The seller never pays a single paisa above the statutory 5% commission cap[5]. On a ₹1,00,000 hammer price, the fisherman receives **₹95,000 (95.0%)** settled directly to their bank account via instant split settlement, plus **₹2,000 (2.0% — representing 40% of the 5% commission)** credited automatically into an auditable **Annual Production Bonus Escrow Ledger**[5]. Fishermen embrace HarbourOS because digital scale locking and live bid logs eliminate weight manipulation, delayed payments, and informal deductions[1].
2. **Licensed Auctioneers Retain Their Full 20% Share (1.0% of Gross)**: The licensed auctioneer receives their full statutory **₹1,000 (20% of the ₹5,000 commission)** automatically upon buyer payment clearing[5]. Auctioneers support the platform because it eliminates manual clerk bookkeeping, enforces their exclusive rights on society-loan vessels under Section 4(8), and shields them from buyer payment defaults[1].
3. **Revenue Stream 1 — B2B Institutional Revenue Share from HMS & Society Allocations**: Under a formal B2B SaaS and Technology Infrastructure Agreement with the Harbour Management Society (HMS/FLCMS) and the participating Co-operative Society, HarbourOS charges a base monthly SaaS subscription (e.g., ₹3,000–₹30,000/month depending on landing centre or harbour size)[1] **plus** a configurable B2B technology share deducted solely from the institutional shares of the HMS and Co-op Society (for example, **20% of the HMS's 1% share = 20 bps of GMV**, and **20% of the Society's 1% share = 20 bps of GMV**)[7]. HMS and Co-op leaders enthusiastically back this arrangement because digitizing weighbridge and auction records captures previously unrecorded transactions, increasing their net treasury receipts even after paying HarbourOS's 20 bps technology split[1].
4. **Revenue Stream 2 — Buyer-Side Digital Clearing, Credit Escrow & Remote Bidding Cut**: Following the proven European **PEFA** marketplace precedent—where local physical participation remains frictionless while remote marketplace access carries a platform fee

(historically reported at 2% for purchasers and 0.2% for member auctions in PEFA's early virtual exchange rollout)[18]—HarbourOS levies a configurable **Buyer Digital Clearing & Escrow Fee (e.g., 50 to 150 bps; 1.00% in our baseline model, plus 18% GST on the service fee)** on wholesale buyers who use remote bidding, digital credit lines, or automated export traceability documentation[7].

5. **Revenue Stream 3 — External Harbour Website SDK & API Gateway Cut**: When a harbour, cooperative federation, or international fish market already has its own website or ERP, they embed the **@harbouros/auction-sdk** and connect via our REST, WebSocket, and Webhook APIs. Even when the auction UI is hosted on the client's domain, checkout and split clearing run through HarbourOS's **Razorpay Route** (or global Stripe Connect) master aggregator pipeline[6]. HarbourOS automatically sweeps its configured **SDK/API Infrastructure Cut (e.g., 25 to 75 bps of GMV)** before releasing the remaining funds to the harbour's linked accounts[6].

## 2.3 Paise-Verified Mathematical Breakdown of a ₹1,00,000 Lot Sale

The table below demonstrates the exact integer-paise conservation of money (1,00,00,000 paise = ₹1,00,000.00 Hammer Price) executed by HarbourOS's settlement engine for a standard Kerala Fishing Harbour auction under Section 4(5)(a) of Act 16 of 2021[5]:

| Ledger Flow Stage | Stakeholder / Account Bucket | Statutory / Contractual Rule | Basis Points (of Hammer Price) | Exact Amount (INR) | Exact Amount (Paise) | Settlement Mechanism |
|---|---|---|---|---|---|---|
| **1. Buyer Inflow** | **Gross Hammer Price** | Winning Auction Bid[1] | 10,000 bps (100.00%) | ₹1,00,000.00 | 10,000,000 | Captured via UPI / NEFT / Escrow Wallet |
| **1. Buyer Inflow** | **HarbourOS Buyer Clearing / SDK Fee** | Configurable Buyer-Side Fee (e.g., 1.0%)[7] | +100 bps (1.00%) | ₹1,000.00 | 100,000 | Retained in HarbourOS Master Account |
| **1. Buyer Inflow** | **GST on Platform Service** | 18% GST on Total Platform | +18 bps (on Buyer Fee) | ₹180.00 | 18,000 | Routed to Tax Liability |

| | | | | | | |
|---|---|---|---|---|---|---|
| | **Fee** | Cut (₹1,400)[7] | | | | Account |
| **Total Inflow** | **Buyer Total Checkout Payable** | **Hammer + Buyer Fee + Buyer Fee GST** | **10,118 bps (101.18%)** | **₹1,01,180.00** | **10,118,000** | **Total Debit from Buyer Account**[7] |
| **2. Seller Payout** | **Fisherman / Vessel Owner Net** | 95% of Hammer Price (5% max deduction)[5] | 9,500 bps (95.00%) | ₹95,000.00 | 9,500,000 | Instant T+0 Razorpay Route Transfer[6] |
| **3. Statutory 5% Split** | **Fisherman Annual Bonus Escrow** | 40% of 5% Commission (Sec 4(5)(a)(iii))[5] | 200 bps (2.00%) | ₹2,000.00 | 200,000 | Credited to Fisher Bonus Escrow Account[7] |
| **3. Statutory 5% Split** | **Licensed Auctioneer Net Payout** | 20% of 5% Commission (Sec 4(5)(a)(i))[5] | 100 bps (1.00%) | ₹1,000.00 | 100,000 | Instant T+0 Razorpay Route Transfer[6] |
| **3. Statutory 5% Split** | **HMS / FLCMS Net Payout** | 20% of 5% (₹1,000) minus 20% B2B Tech Share[5] | 80 bps (0.80%) | ₹800.00 | 80,000 | Razorpay Route Transfer to HMS Treasury[7] |
| **3. Statutory 5%** | **Co-operative Society Net** | 20% of 5% (₹1,000) minus | 80 bps (0.80%) | ₹800.00 | 80,000 | Razorpay Route Transfer to Co-op |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Split** | **Payout** | 20% B2B Tech Share[5] | | | | Society[7] |
| **4. Platform Cut** | **Harbour OS Total Gross Revenue** | **40 bps B2B Split (₹400) + 100 bps Buyer Fee (₹1,000)**[7] | **140 bps (1.40%)** | **₹1,400.00** | **140,000** | **Retained in Harbour OS Master Account**[7] |

## 2.4 Variable Commission Configuration Matrix Across Markets

Because every harbour and jurisdiction operates under different bylaws, the **Super Admin Control Plane** assigns a versioned, effective-dated CommissionPolicy per tenant:

| **Jurisdiction / Deployment Mode** | **Seller Commission Cap** | **Statutory / Local Apportionment Rule** | **HarbourOS B2B Cut (Variable bps)** | **HarbourOS Buyer / SDK Cut (Variable bps)** | **Settlement Rail & Compliance** |
|---|---|---|---|---|---|
| **Kerala Fishing Harbour / Landing Centre** (Hosted SaaS)[5] | **5.00%** (500 bps) under Act 16 Sec 4(4)[5] | 20% Auctioneer, 20% HMS/FLCMS, 40% Fisher Bonus Escrow, 20% Co-op Society[5] | **10–30 bps** (Agreed B2B share of HMS & Society allocations)[7] | **50–150 bps** (Buyer Clearing & Credit Escrow Fee)[7] | Razorpay Route (Requires FY25/FY26 ₹40L domestic or ₹5L export turnover per linked entity or institutional aggregator setup)[6] |
| **Kerala Public / LSGI Fish Market** (Hosted | **5.00%** (500 bps) under Act 16 Sec 4(5)(b)[5] | 20% Auctioneer, 20% FMMS, 30% Welfare | **15–25 bps** (Agreed B2B share of FMMS & LSGI | **50–100 bps** (Wholesale Buyer Clearing | Razorpay Route + Government Treasury e-Challan |

| | | | | | |
|---|---|---|---|---|---|
| SaaS)[5] | | Fund Board, 30% LSGI/Owner[5] | allocations) | Fee) | Remittance[6] |
| **Pan-India Cooperative / Private Harbour** (Gujarat, AP, TN, Maharashtra, Odisha) | **2.00%–6.00%** (Configurable per society bylaws) | Configurable N-way split between Vessel Owner, Society, and Auctioneer | **20–50 bps** of GMV or tiered monthly SaaS subscription[1] | **50–150 bps** (Remote Buyer & B2B Commerce Fee) | Razorpay Route / Cashfree Split / ONDC Network[1] |
| **External Website / ERP via HarbourOS SDK** (India & Global)[1] | Defined by Host Harbour | Host Harbour configures its local split rules via API | **25–75 bps** API/SDK Infrastructure & Escrow Cut | **50–200 bps** Remote Network Buyer Fee (PEFA model)[18] | Razorpay Route (India) / Stripe Connect & Adyen Marketplaces (Global) |

# 3. Frictionless Onboarding & Role-Based Mobile-First UI/UX Plan

Harbour operations take place in harsh physical conditions: workers have wet, fish-scaled hands, operate under pre-dawn darkness or blinding coastal sun, face intermittent cellular connectivity, and include stakeholders with limited digital literacy[1]. HarbourOS enforces a **"Zero-Typing, High-Contrast, Offline-First"** UX standard across six role-specific interfaces[1]:

## 3.1 Smooth, Assisted Multi-Role Onboarding Workflow

1. **Fishermen & Boat Owners (60-Second Assisted Kiosk Onboarding — Smartphone Optional)**:
   - Fishermen do not need to fill out long forms or even own a smartphone to participate[1]. During a 1-day harbour onboarding camp, a harbour clerk scans the fisherman's **NFDP Work-Based Identity Card**[11] or enters their **ReALCraft Vessel Registration Number** and mobile number[1].
   - The system links their bank account via instant ₹1 UPI/IMPS penny-drop verification and records whether the vessel has an active **Fishermen Development Welfare Co-operative Society equipment loan** (automatically enforcing Section 4(8) so only society-deputed auctioneers can conduct their auctions)[5].

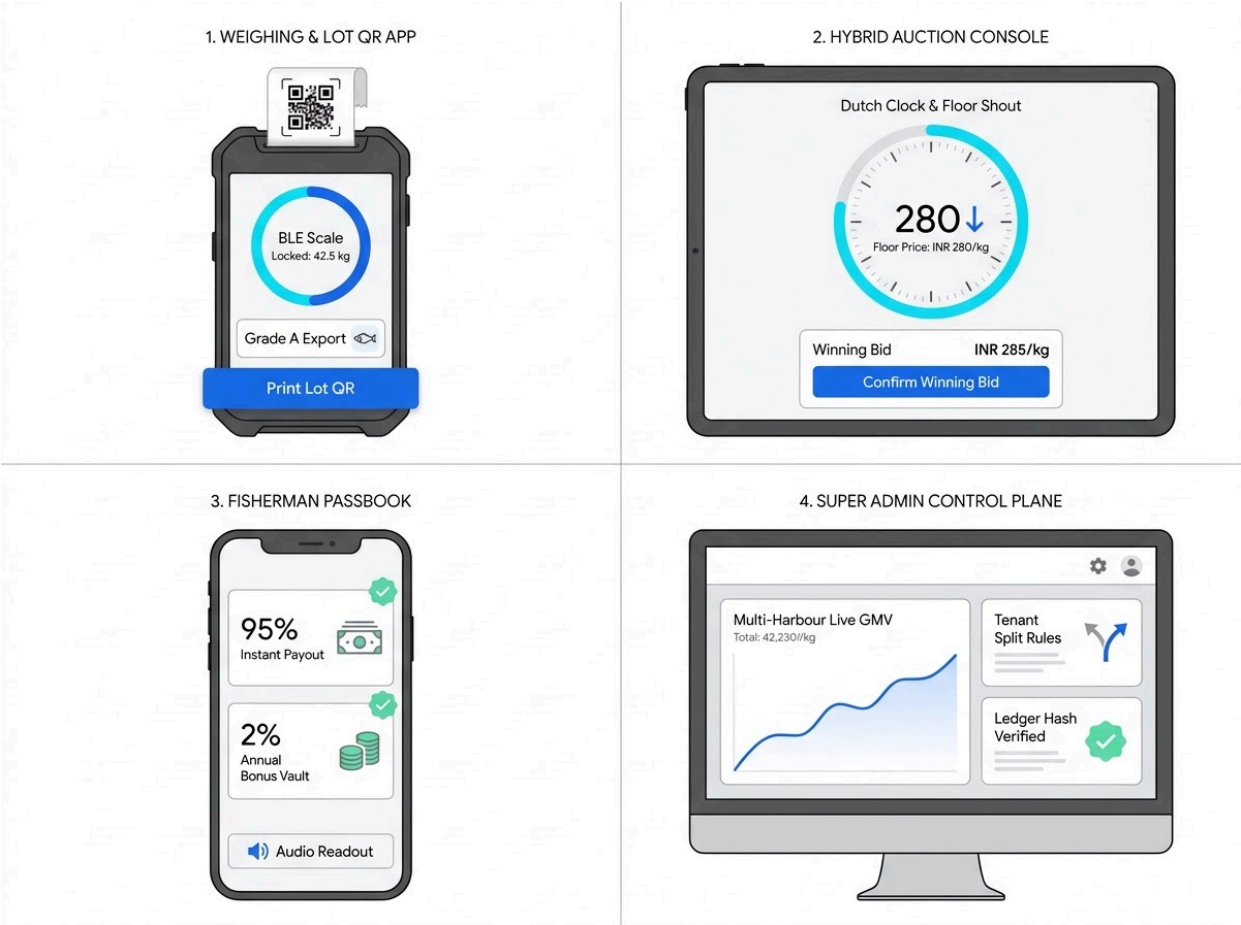
- Fishermen without smartphones receive a waterproof, laminated **PVC QR Smart Passbook Card** and receive automated Malayalam SMS/WhatsApp receipts and thermal printed slips after every auction[1].

2. **Wholesale Buyers & Exporters (Digital Bank Guarantee & Unified Buyer Number)**:
   - Inspired by **PEFA's** proven onboarding flow—where a buyer provides a bank guarantee to the auction, receives a unique buyer number, and links it to their trading subscription[14]—wholesale buyers onboard via GSTIN/PAN verification and post either a digital escrow deposit, bank guarantee reference, or UPI AutoPay mandate.
   - The buyer is issued a unified **HarbourOS Buyer Number** that works across all authorized harbours, with real-time dynamic credit enforcement (Available Bidding Limit = Escrow Balance + Harbour Approved Credit Limit - Unsettled Payables)[1].
3. **Harbour Tenants & External SDK Partners (10-Minute Self-Serve Provisioning)**:
   - Harbour Management Societies or external portal operators onboard via a step-by-step wizard: selecting their regulatory preset (e.g., KERALA_ACT_16_HARBOUR), uploading licensed auctioneer rosters with 3-year expiry tracking under Section 5[5], completing Razorpay Route linked-account KYC (including GST-3B or FIRC verification where required by RBI Payment Aggregator rules)[6], and generating publishable/secret keys for external SDK embedding.

## 3.2 Role-by-Role Interface & Dashboard Design

- **1. Weighing & Grading Operator App (Mobile-First Rugged Android — Malayalam & English)**
  - **Ergonomics**: One-handed thumb layout with oversized buttons (min-height: 64px), high-contrast Daylight Mode (#FFFFFF surface, #111111 text) and Pre-Dawn Dark Mode (#0B0F19 surface).
  - **Workflow**: Operator scans the Boat's QR card -> taps the **Visual Fish Species Grid** (displaying high-contrast photos alongside Malayalam and English names such as *Mathi / Sardine*, *Ayala / Mackerel*, *Neymeen / Seer Fish*, *Choora / Tuna*) -> weight locks automatically from the **Bluetooth Low Energy (BLE) Digital Scale** (manual weight overrides require a supervisor PIN and log an immutable audit event)[1] -> taps Quality Grade (Grade A Export, Grade A Domestic, Grade B Processing aligned with Section 21 certification)[5] -> Bluetooth thermal printer outputs a waterproof **Lot QR Crate Sticker** in <1.5 seconds[1].
- **2. Auctioneer Hybrid Floor & Clock Console (Tablet & Web)**
  - **Ergonomics**: Combines **Aucxis KOSMOS's** multi-method sales flexibility (falling-price Dutch clock, rising price, and tablet shout registration)[3] with **Toyosu's** rapid floor execution[15].
  - **Workflow**: Auctioneer scans the Lot QR sticker to open bidding. The console supports three selectable modes:
    - *Physical Shout Keypad Mode*: Auctioneer conducts traditional vocal or hand-signal bidding on the shed floor[10] and taps quick price increments (+₹10, +₹50, +₹100, +₹500) plus the winning buyer's 3-digit paddle number[1].

    - *Hybrid Floor + Remote Live Mode*: Floor bids entered on the auctioneer's tablet compete in real time (<50ms WebSocket sync) against remote wholesale buyers bidding from their mobile/web terminals[3].
    - *Dutch Descending Clock Mode*: The radial clock starts at an upper price and ticks downward; the first floor or remote buyer to press "Stop Clock" wins the lot and selects the number of boxes required[3].
  - **Built-In Statutory Guardrails**: The console automatically rejects any bid below the HMS-configured **Section 19 Basic Floor Price**[5] or from a buyer whose credit limit is exceeded[1].
- **3. Fisherman Mobile App & Voice Passbook (Android PWA / Native)**
  - **Ergonomics**: Icon-driven interface with a prominent **"Tap-to-Listen in Malayalam" (മലയാളത്തിൽ കേൾക്കുക)** voice readout button on every screen for fishermen with limited reading literacy[1].
  - **Workflow**: Live push alerts when catch is weighed, auctioned, and settled[1]; clear visual breakdown showing the **95% Instant Payout** and the accumulating **2% Annual Production Bonus Vault** (under Section 4(5)(a))[5]; cross-harbour live species price discovery[1]; and downloadable settlement statements for NFDP bank credit applications[11].
- **4. Wholesale Buyer Terminal & Exporter Desk (Mobile App + Multi-Screen Web)**
  - **Workflow**: Watch up to 4 live harbour auctions simultaneously[14]; view high-resolution catch photos (including tail-cut cross-sections for tuna and export species, inspired by Toyosu and PEFA quality grading)[14]; place one-tap remote bids; settle invoices via UPI/NEFT/Credit Line; track cold-chain dispatch vehicles; and generate one-click **MPEDA EU Catch Certificate (EC Regulation 1005/2008) and US DS-2031 QR Traceability packets**[19].
- **5. Harbour Administrator, HMS & Accountant Portal (Web)**
  - **Workflow**: Publish daily **Section 19 Basic Floor Prices**[5]; manage auctioneer licences and society loan tie-ups[5]; monitor buyer credit aging and bank guarantees[1]; review and dual-sign offline synchronization conflicts or financial adjustments[1]; and export daily/monthly GST, treasury, and society apportionment reports in PDF and Excel[1].
- **6. Platform Super Admin Command Center ("God-Mode" Multi-Tenant Control Plane)**
  - **Workflow**: Real-time observability across all harbours, landing centres, and external SDK partner sites[1]; tenant onboarding and custom domain provisioning; **Per-Harbour Variable Commission & Split Rule Configurator** (adjusting Seller Cap bps, 4-way Statutory Apportionment bps, HarbourOS B2B Share bps, and Buyer/SDK Clearing Fee bps with versioned audit history)[5]; **External SDK & API Telemetry Monitor** (tracking API key usage, webhook delivery latency, and SDK checkout conversion); **Global Cryptographic Ledger Verifier** (continuous background verification of every tenant's SHA-256 hash chain); and automated tenant suspension switches for overdue accounts (mirroring PEFA's 3-step reminder, 7-day notice, and suspension workflow)[2].




# 4. Development Budget, Pilot Costs & Illustrative Revenue Projections

## 4.1 Verified Indian Development & Pilot Cost Benchmarks

Based on Indian software engineering and harbour hardware planning estimates[1], launching the Phase 1 MVP and a single-harbour live pilot requires a total capital budget of **₹20–32 lakh** (which can be reduced to **₹1–3 lakh** in direct cash outlay if the founding engineering team builds the software in-house and utilizes existing devices)[1]:

- **Custom Software Engineering Budget (₹15–24 Lakh Total)**: Requirements and field research (₹1–2 lakh), UI/UX design (₹1–1.5 lakh), Backend development (₹4–6 lakh), Web application (₹3–4 lakh), Mobile application (₹3–5 lakh), Database and integrations (₹1.5–2.5 lakh), Testing and security (₹1–2 lakh), and Deployment and documentation (₹0.5–1 lakh)[1].
- **Dockside Hardware & On-Site Pilot Budget (₹1.75–3.75 Lakh Total per Pilot Harbour)**: Digital weighing scale integration (₹30,000–80,000), rugged tablets/smartphones (₹40,000–80,000), thermal QR printer and labels (₹15,000–35,000), local networking and connectivity (₹15,000–30,000), on-site installation (₹20,000–40,000), user training (₹25,000–50,000), and operational

contingency (₹30,000–60,000)[1].

- **Monthly Cloud & Messaging Operating Costs (₹19,000–53,000/month)**: Cloud hosting and managed database (₹5,000–15,000), backups and monitoring (₹1,000–3,000), SMS and WhatsApp Business API notifications (₹1,000–5,000, subject to AWS and Meta per-message tariffs), maintenance and support (₹10,000–25,000), and miscellaneous services (₹2,000–5,000)[1].

## 4.2 Illustrative Commercial Scaling: Pure SaaS vs. Hybrid Clearing Cut

In our baseline commercial study, fixed monthly SaaS subscriptions are priced at **₹3,000–5,000/month** for small landing centres, **₹8,000–15,000/month** for medium harbours, and **₹15,000–30,000/month** for large harbours[1]. At an illustrative average subscription of **₹10,000/site/month**, pure subscription Annual Recurring Revenue (ARR) equals **₹6 lakh at 5 sites**, **₹18 lakh at 15 sites**, and **₹36 lakh at 30 sites**[1].

When we layer our proposed **Variable Clearing Cut** (e.g., an illustrative **30 bps B2B Institutional Share + 70 bps Buyer/SDK Clearing Fee = 1.00% combined platform take-rate** on an illustrative ₹5 Crore/month gross auction volume per medium harbour) onto the base SaaS subscription, the illustrative revenue trajectory expands dramatically[1]:

| **Commercial Scenario (Illustrative Planning Model)** | **5 Active Sites (Pilot Cluster)** | **15 Active Sites (Kerala Scale)** | **30 Active Sites (National + SDK Expansion)** |
|---|---|---|---|
| **Pure SaaS Subscription ARR** (at ₹10,000/month/site baseline)[1] | **₹6.0 Lakh / year** (₹0.5L/mo)[1] | **₹18.0 Lakh / year** (₹1.5L/mo)[1] | **₹36.0 Lakh / year** (₹3.0L/mo)[1] |
| **Illustrative Monthly GMV** (Assuming ₹5 Cr/month/site) | ₹25.0 Crore / month | ₹75.0 Crore / month | ₹150.0 Crore / month |
| **Illustrative Clearing & SDK Cut ARR** (at 100 bps combined B2B + Buyer Cut)[7] | ₹3.0 Crore / year (₹25L/mo) | ₹9.0 Crore / year (₹75L/mo) | ₹18.0 Crore / year (₹1.50 Cr/mo) |
| **Combined Hybrid ARR (SaaS +** | **₹3.06 Crore / year** | **₹9.18 Crore / year** | **₹18.36 Crore / year** |

| | | | |
|---|---|---|---|
| **Clearing Cut)** | | | |

# FILE 2: HARBOUROS_SRS_AND_BUILD_PLAN.md

## 1. Software Requirements Specification (SRS) — Production Baseline

### 1.1 Purpose, Scope & Definitions

This Software Requirements Specification (SRS) defines the complete functional, non-functional, database, hardware, and external SDK/API requirements for **HarbourOS**[1]. It expands the Version 1.0 baseline (FR-001 through FR-083, NFR-001 through NFR-024, and BR-001 through BR-009)[1] into a zero-ambiguity microservices specification covering Kerala Act 16 of 2021 statutory apportionment, hash-chained double-entry accounting, and third-party website SDK federation[5].

### 1.2 Complete Functional Requirements (FR)

#### 1.2.1 Authentication, Access Control & Statutory Licensing (svc-identity)

- **FR-001 to FR-006 (Multi-Tenant RBAC, OTP Auth & Security Audit)**: The system shall support authorized user registration (FR-001), passwordless login via mobile number + OTP with Malayalam voice-call fallback and hardware biometric/PIN token binding (FR-002), and strict Role-Based Access Control across eight user classes: SUPER_ADMIN, HARBOUR_ADMIN, LICENSED_AUCTIONEER, FISHERMAN_BOAT_OWNER, WHOLESALE_BUYER, WEIGHING_OPERATOR, ACCOUNTANT, and AUTHORIZED_INSPECTOR (FR-003)[1]. Administrators can activate, suspend, or deactivate accounts (FR-004), access is strictly scoped to assigned role and harbour tenant (FR-005), and all login and security events are logged (FR-006)[1].
- **FR-084 (Section 5 Auctioneer Licence Validation)**: The system shall record each auctioneer's statutory licence number, issuing Authorised Officer reference, security deposit reference, permitted harbour_id, and 3-year validity window (valid_from, valid_until) per Section 5 of Kerala Act 16 of 2021[5]. The system shall reject any attempt to create or conduct an auction if the licence is expired, suspended, or invoked outside the permitted harbour (BR-001, BR-002)[1].

#### 1.2.2 Multi-Harbour Tenant, Fleet & Buyer Management (svc-harbour)

- **FR-007 to FR-010 (Harbour Tenant Provisioning & Settings)**: The Super Administrator shall create and configure independent harbour tenants (FR-007), isolating each harbour's users and operational records (FR-008) while enabling the Harbour Administrator to manage harbour-specific location, contact, and operational

configurations (FR-009, FR-010)[1].

- **FR-011 to FR-017 (Fisherman, Vessel & Co-op Loan Registry)**: Authorized staff shall register fishermen (FR-011) with contact and external government ID references (nfdp_registration_no, realcraft_vessel_id, self_permit_id) without altering government records (FR-012, FR-014)[1], associate multiple fishermen with vessels (FR-013, FR-015), update profiles (FR-016), and preserve historical vessel-ownership associations (FR-017)[1].
- **FR-085 (Section 4(8) Co-op Society Loan Encumbrance Enforcement)**: Each vessel record shall track has_active_society_loan and loan_coop_society_id[5]. When has_active_society_loan = TRUE, the system shall enforce Section 4(8) of Kerala Act 16 of 2021 by restricting auction assignment for that vessel's lots exclusively to licensed auctioneers whose profile has a matching deputed_by_society_id[5].
- **FR-018 to FR-022 (Wholesale Buyer & Credit Limit Management)**: Authorized staff shall register wholesale buyers (FR-018), store business/GST and bank guarantee details (FR-019)[1], verify and activate buyer accounts (FR-020), and maintain real-time buyer transaction history (FR-021) and outstanding credit/escrow balances (FR-022)[1].
- **FR-086 (Section 19 Basic Floor Price Schedule)**: The Harbour Administrator (on behalf of the HMS) shall configure species-wise and quality-grade-wise **Basic Floor Prices** (basic_floor_price_paise) with effective timestamps per Section 19 of Act 16 of 2021[5].

### 1.2.3 Catch Registration, BLE Weighing & Lot Serialization (svc-catch-lot)

- **FR-023 to FR-029 (Fishing Trip & Catch Arrival Registration)**: Authorized users shall create catch records (FR-023) linked to a vessel and fishing trip (FR-024), capturing arrival date, time, and landing centre (FR-025), supporting multiple fish species per catch (FR-026), recording estimated and verified quantities (FR-027), attaching photos and quality certificates (FR-028), and maintaining a historical catch register (FR-029)[1].
- **FR-030 to FR-038 (Fish Lot Creation, BLE Weighing & QR Labeling)**: Authorized operators shall create individual fish lots (FR-030) with globally unique ULID/UUIDv7 identifiers (FR-031) linked to a catch record (FR-032), recording species, net weight, quality grade, and box/basket packaging count per Section 4(7) (FR-033, FR-034)[1]. The system shall support both manual weight entry (FR-035) and direct hardware integration with compatible Bluetooth Low Energy (BLE) / RS-232 digital weighing scales (FR-036), require authorization and an immutable audit log for any weight correction (FR-037), and print a signed QR code sticker for every lot (FR-038)[1].

### 1.2.4 Hybrid Real-Time Auction Engine & External SDK (svc-auction-engine & svc-sdk-gateway)

- **FR-039 to FR-050 (Multi-Mode Auction Execution)**: Licensed auctioneers shall create auctions containing one or more fish lots (FR-039, FR-040) and manage auction scheduling and status (FR-041)[1]. The engine shall support four execution modes[3]:

1. **Physical Shout Fast-Entry (FR-042)**: Rapid tablet recording of physical floor bids[1].
2. **Electronic & Hybrid Live Bidding (FR-043)**: Simultaneous physical floor bidding and remote WebSocket bidding[1].
3. **Dutch Descending Clock (FR-087)**: Automated high-to-low clock ticker where the first

buyer to stop the clock wins the lot or partial box quantity (PEFA / Aucxis KOSMOS standard)[3].

4. **Co-op Basic Price Procurement (FR-088)**: Direct procurement by the Fishermen Development Welfare Co-operative Society at the HMS-fixed Section 19 basic price, or HMS direct first-sale takeover under Section 10[5].

- **FR-044 to FR-050 (Bid Validation, Audit & Immutability)**: Every bid shall record buyer_id, lot_id, amount_paise, server nanosecond timestamp, and submitting_user_id (FR-044), preserving the complete bid history (FR-045)[1]. The engine shall reject any bid below the Section 19 basic floor price or exceeding the buyer's available credit limit, identify the winning bid (FR-046), allow authorized sale confirmation or cancellation with recorded reason codes (FR-047, FR-050), block unauthorized edits to completed auctions (FR-048), and emit an immutable auction result record (FR-049)[1].
- **FR-089 (External Website SDK & Webhook Federation)**: The platform shall provide @harbouros/auction-sdk (Web Components/React/Flutter) and authenticated REST/WebSocket/Webhook APIs so third-party harbour portals can embed live HarbourOS auctions or sync external auction confirmations into HarbourOS's clearing engine while automatically deducting HarbourOS's configured SDK/API fee cut.

### 1.2.5 Sales, Tamper-Proof Double-Entry Ledger & Split Settlement (svc-ledger-settlement)

- **FR-051 to FR-057 (Sale Transaction & Commission Apportionment)**: Confirming an auction shall atomically create a sale transaction (FR-051) linking lot, buyer, seller, quantity, and final hammer price (FR-052), preventing duplicate sales for the same lot via a database unique constraint (FR-057, BR-005)[1]. The system shall calculate gross value (FR-053), compute seller commission strictly within the tenant's statutory cap (<= 5% in Kerala under Act 16 Section 4(4)) (FR-054, BR-003), apportion the commission across the statutory buckets (20% Auctioneer, 20% HMS/FLCMS, 40% Fisherman Annual Bonus Escrow, 20% Co-op Society at harbours; or 20/20/30/30 at fish markets)[5], compute HarbourOS's variable B2B share and Buyer/SDK clearing fee (FR-055)[7], and generate a bilingual digital/printed receipt (FR-056)[1].
- **FR-058 to FR-065 (Double-Entry Ledger, Escrow & Razorpay Route Settlement)**: The system shall track payment status (PENDING, PARTIAL, COMPLETED) independently from auction completion (FR-058, FR-060, BR-006), supporting cash, bank transfer, UPI, and Razorpay Route / Cashfree split checkout (FR-059)[1]. The system shall calculate net fisherman payouts (FR-061), track buyer credit balances (FR-062), generate fisherman settlement statements (FR-063), enable accountant reconciliation (FR-064), and enforce a **cryptographic SHA-256 hash-chained, append-only double-entry ledger** where zero financial records can ever be updated or deleted (BR-007) and any adjustment requires dual-role authorization (HARBOUR_ADMIN + ACCOUNTANT) via a compensating reversal journal (FR-065, BR-008)[1].

### 1.2.6 Reporting, MPEDA Traceability, Notifications & Offline Sync

- **FR-066 to FR-073 (Operational, Financial & Export Traceability Reports)**: Generate

daily auction reports (FR-066), monthly sales reports (FR-067), fisherman earnings (FR-068), buyer purchases (FR-069), species-wise volume and historical price trends (FR-070, FR-071), outstanding buyer credit aging (FR-072), and PDF/Excel exports (FR-073)[1], plus **MPEDA EU Catch Certificate (Regulation 1005/2008), Japan Catch Documentation, and US DS-2031 QR Traceability data packets** (FR-090)[19].

- **FR-074 to FR-077 (Multilingual Notifications)**: Dispatch real-time SMS, WhatsApp Business API, and push notifications in Malayalam and English for auction events (FR-074), buyer winning bids/invoices (FR-075), and fisherman settlement payouts (FR-076, FR-077)[1].
- **FR-078 to FR-083 (Offline-First Edge Synchronization)**: Store authorized dockside records locally in encrypted SQLite (FR-078), synchronize automatically upon connectivity restoration (FR-079), flag sync errors (FR-080), enforce UUIDv7 idempotency keys to prevent duplicate transactions (FR-081), quarantine conflicting updates for supervisor resolution (FR-082), and strictly forbid overwriting completed auction or payment records (FR-083)[1].

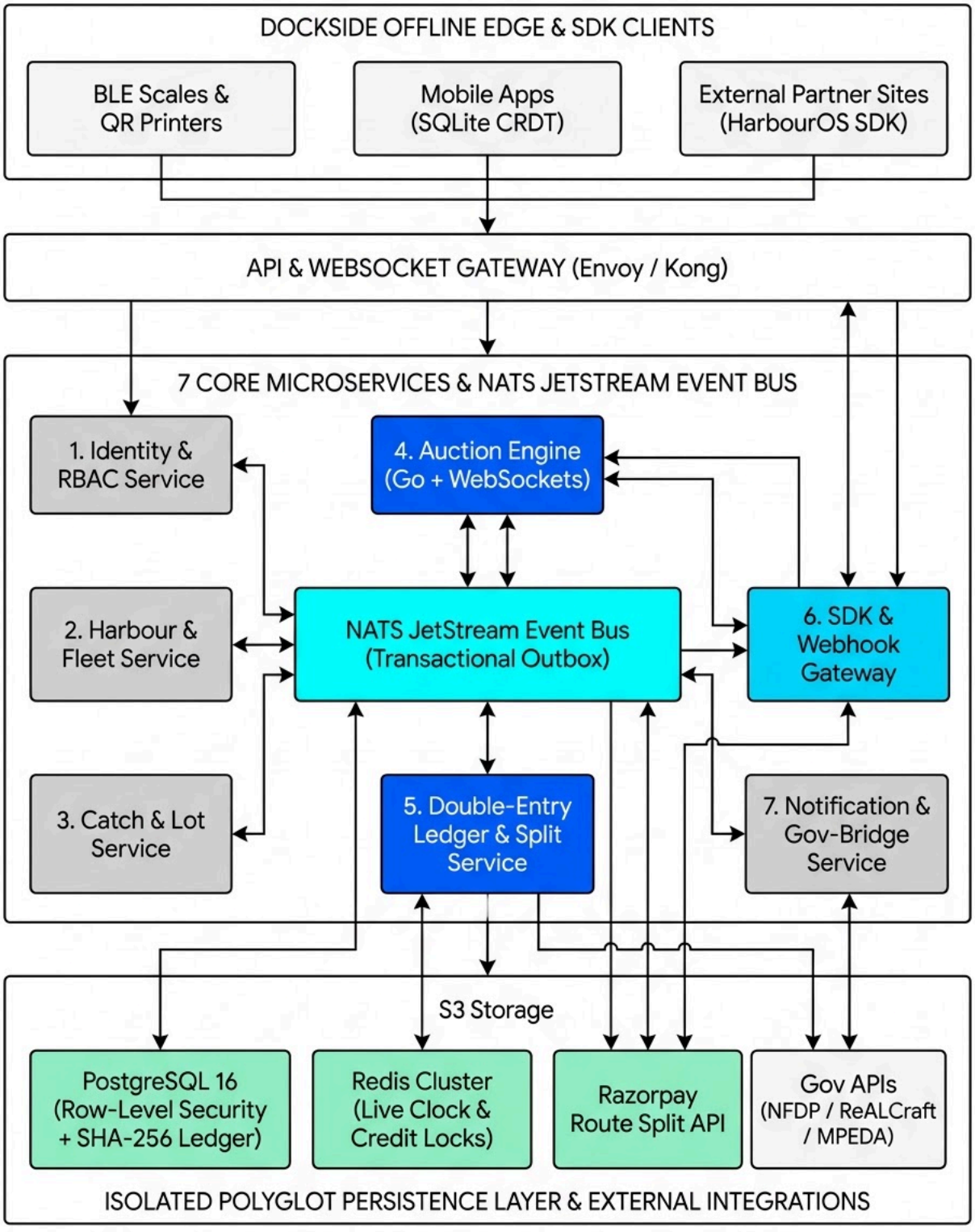
### 1.3 Non-Functional Requirements (NFR)

- **Performance (NFR-001 to NFR-004)**: Page loads <3 seconds on pilot 4G networks (NFR-001); standard DB queries <2 seconds (NFR-002); support >=100 concurrent users per single-harbour pilot pod and scale horizontally to 10,000+ concurrent bidders across multi-harbour clusters (NFR-003); sub-50ms WebSocket bid ordering with nanosecond server timestamps (NFR-004)[1].
- **Security (NFR-005 to NFR-012)**: TLS 1.3 / HTTPS everywhere (NFR-005); Argon2id secret hashing and hardware-bound JWTs (NFR-006); RBAC + PostgreSQL Row-Level Security tenant isolation (NFR-007, NFR-023); AES-256 encryption at rest (NFR-008); SHA-256 hash-chained ledger audit logs (NFR-009, NFR-011); automated point-in-time recovery (PITR) backups (NFR-010); and OWASP Top 10 / rate-limiting protection at the API Gateway (NFR-012)[1].
- **Availability, Usability & Scalability (NFR-013 to NFR-024)**: Target 99.5% to 99.99% cloud uptime outside scheduled non-auction maintenance windows (NFR-013)[1]; 100% offline dockside continuity (NFR-014, NFR-016); bilingual Malayalam and English UI optimized for low-cost Android devices (NFR-017 to NFR-021); and stateless horizontal pod autoscaling (NFR-022, NFR-024)[1].

# 2. Pre-Refactored Microservices Architecture & Workspace Layout

While a small single-site prototype can start as a modular monolith[1], your requirement for a **multi-tenant, multi-harbour, horizontally scalable SaaS with an external SDK and zero refactoring nightmares** demands a **Pre-Refactored Domain-Isolated Microservices Architecture inside a Unified Nx Monorepo**. By establishing strict service boundaries, Protobuf gRPC contracts, and isolated database schemas inside a single monorepo from day one, you get the clean deployment ergonomics of a unified codebase alongside the

independent horizontal scaling of microservices.




## 2.1 Microservice Decomposition & Horizontal Scaling Matrix

| Service / Package Path | Runtime & Tech Stack | Isolated DB Schema & State | Core Domain Responsibilities & Horizontal Scaling Rule |
|---|---|---|---|
| **apps/svc-identity** | Node.js 22 + NestJS (Fastify) + gRPC | schema: identity + Redis Session Store | Multi-tenant user registration, OTP/Voice auth, WebAuthn device binding, RBAC policy evaluation, and Section 5 Auctioneer Licence registry[1]. Stateless HPA scaling (2–10 pods). |
| **apps/svc-harbour** | Node.js 22 + NestJS (Fastify) + gRPC | schema: harbour | Tenant provisioning, Harbour/Landing Centre config, Section 19 Basic Floor Price schedules, Vessel/Fisherman mapping, and Section 4(8) Co-op Society loan encumbrance rules[1]. Stateless HPA scaling (2–8 pods). |
| **apps/svc-catch-lot** | Node.js 22 + NestJS (Fastify) + gRPC | schema: catch_lot + S3 Object Storage | Fishing trip arrival logs, BLE scale weight ingestion, tare deduction, Section 21 quality grading, Lot ULID/QR generation, and offline CRDT batch sync ingestion[1]. |

| | | | |
|---|---|---|---|
| | | | Stateless HPA scaling (2–12 pods). |
| **apps/svc-auction-engine** | **Go 1.23** (Gorilla WS / Fiber + gRPC) | schema: auction + **Redis 7 Cluster** | Sub-50ms Dutch Descending Clock ticker, Hybrid Floor + Remote WebSocket rooms, atomic buyer credit pre-checks, Redlock lot locking, and SaleConfirmed event emission[3]. Partitioned horizontally by harbour_id across Redis Pub/Sub shards (4–30 pods). |
| **apps/svc-ledger-settlement** | Node.js 22 + NestJS (Fastify) + NATS | schema: ledger (Strict SERIALIZABLE / Row-Locked) | Append-only SHA-256 hash-chained double-entry ledger, Kerala Act 16 4-way statutory split calculation, HarbourOS B2B + Buyer/SDK cut calculation, and Razorpay Route / Stripe Connect split payouts[5]. Scales via partitioned NATS JetStream consumer groups (3–15 pods). |
| **apps/svc-sdk-gateway** | Node.js 22 + NestJS (Fastify) | schema: sdk_integrations + Redis Rate-Limiter | Public REST/WebSocket gateway for @harbouros/auction-sdk, OAuth2 |

| | | | |
|---|---|---|---|
| | | | client-credentials & API key validation, idempotency enforcement, and exponential-backoff HMAC webhook delivery. |
| **apps/svc-gov-notify** | Node.js 22 + NestJS (Fastify) | schema: notifications | Bilingual Malayalam/English WhatsApp Business API & SMS notifications[1], PDF/Excel report generator, MPEDA EU Catch Certificate / DS-2031 QR packet builder[19], and async adapters for NFDP/ReALCraft[1]. |
| **packages/auction-sdk-js & packages/domain-money** | TypeScript (Zero-Dependency Shared Libraries) | N/A (Stateless Libraries) | @harbouros/auction-sdk provides drop-in Web Components/React widgets for external harbour sites; domain-money enforces BigInt paise & basis-point Largest-Remainder split math[7]. |

# 3. Database Connection Architecture & Tamper-Proof PostgreSQL Schema

## 3.1 Multi-Tenant Database Connection Pooling & Row-Level Security (RLS)

To prevent microservice pods from exhausting PostgreSQL connection limits during peak 5:30 AM harbour auctions while guaranteeing zero cross-tenant data leakage (NFR-023)[1], all

microservices connect through **PgBouncer** (pool_mode = transaction) using a shared TenantAwareDbPool driver that injects app.current_tenant_id locally into every SQL transaction:

```
TypeScript
// packages/db-rls-driver/src/tenant-db.provider.ts
import { Pool, PoolClient } from 'pg'

export class TenantAwareDbPool {
  private readonly pool: Pool

  constructor(connectionString: string, schemaName: string) {
    this.pool = new Pool({
      connectionString, // Points to PgBouncer port 6432: postgresql://svc_ledger:***@pgbouncer.internal:6432/harbouros_prod
      max: 25,                  // Max connections per microservice pod
      idleTimeoutMillis: 30_000,
      connectionTimeoutMillis: 3_000,
      application_name: `harbouros_${schemaName}`,
      options: `-c search_path=${schemaName},public -c statement_timeout=5000`,
    });
  }

  /**
   * Executes a callback inside an ACID transaction with PostgreSQL Row-Level Security (RLS) strictly locked to the authenticated tenantId and userId.
   */
  async withTenantTransaction<T>(
    tenantId: string,
    userId: string,
    role: string,
    isolationLevel: 'READ COMMITTED' | 'REPEATABLE READ' | 'SERIALIZABLE',
    fn: (client: PoolClient) => Promise<T>,
  ): Promise<T> {
    const client = await this.pool.connect();
    try {
      await client.query(`BEGIN ISOLATION LEVEL ${isolationLevel}`);
      await client.query(
        `SELECT set_config('app.current_tenant_id', $1, true),
                set_config('app.current_user_id', $2, true),
                set_config('app.current_user_role', $3, true)`
```

```
    [tenantId, userId, role]
  )
  const result = await fn(client)
  await client.query('COMMIT')
  return result
 } catch (err) {
  await client.query('ROLLBACK')
  throw err;
 } finally {
  client.release()
 }
}
```

## 3.2 Complete Production PostgreSQL 16 DDL (Multi-Tenant + Immutable Hash-Chain Ledger)

```
SQL
-- ============================================================================
-- HARBOUROS CORE DDL: MULTI-TENANT RLS + CRYPTOGRAPHIC DOUBLE-ENTRY LEDGER
-- ============================================================================
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

CREATE SCHEMA IF NOT EXISTS harbour;
CREATE SCHEMA IF NOT EXISTS catch_lot;
CREATE SCHEMA IF NOT EXISTS auction;
CREATE SCHEMA IF NOT EXISTS ledger;

-- 1. TENANTS & HARBOUR CONFIGURATION (Variable Commission & Regulatory Rules)
CREATE TABLE harbour.tenants (
  tenant_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  slug               VARCHAR(64) UNIQUE NOT NULL,
  name_en            VARCHAR(255) NOT NULL,
  name_ml            VARCHAR(255),
  tenant_type        VARCHAR(32) NOT NULL CHECK (tenant_type IN (
                       'KERALA_HARBOUR', 'KERALA_LANDING_CENTRE', 'KERALA_MARKET',
                       'INDIA_COOP_HARBOUR', 'GLOBAL_EXCHANGE', 'EXTERNAL_SDK_SITE'
                     )),
  deployment_mode    VARCHAR(24) NOT NULL CHECK (deployment_mode IN
```

```
'HOSTED_SAAS', 'EXTERNAL_SDK', 'HYBRID'
    currency_code       CHAR(3) NOT NULL DEFAULT 'INR',
    razorpay_account_id VARCHAR(64),  -- Linked Account for HMS / Harbour Owner
    is_active           BOOLEAN NOT NULL DEFAULT TRUE,
    created_at          TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
);

-- Configurable Variable Commission & Split Policy per Harbour (Versioned & Auditable)
CREATE TABLE harbour.commission_policies (
    policy_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id           UUID NOT NULL REFERENCES harbour.tenants(tenant_id),
    version_no          INT NOT NULL,
    -- Statutory Seller Commission Cap (e.g., 500 bps = 5.00% under Kerala Act 16 Sec 4(4))
    seller_commission_bps INT NOT NULL CHECK (seller_commission_bps BETWEEN 0 AND 1500),
    -- Apportionment of Seller Commission (Must sum to 10,000 bps = 100% of Seller Commission)
    split_auctioneer_bps INT NOT NULL DEFAULT 2000,  -- 20% of 5% = 1.00%
    split_hms_or_market_bps INT NOT NULL DEFAULT 2000,  -- 20% of 5% = 1.00%
    split_fisher_bonus_or_wf_bps INT NOT NULL DEFAULT 4000,  -- 40% of 5% = 2.00% (or 30% Welfare Fund)
    split_coop_or_lsgi_bps INT NOT NULL DEFAULT 2000,  -- 20% of 5% = 1.00% (or 30% LSGI)
    -- HarbourOS Variable B2B Cut (Deducted from HMS & Society shares, never from Fisher/Auctioneer)
    harbouros_cut_from_hms_bps INT NOT NULL DEFAULT 2000,  -- e.g. 20% of HMS share = 20 bps of GMV
    harbouros_cut_from_coop_bps INT NOT NULL DEFAULT 2000,  -- e.g. 20% of Society share = 20 bps of GMV
    -- HarbourOS Variable Buyer-Side Clearing / SDK Cut (Charged on top of Hammer Price)
    buyer_clearing_fee_bps INT NOT NULL DEFAULT 100,  -- e.g. 100 bps = 1.00%
    sdk_external_api_fee_bps INT NOT NULL DEFAULT 50,  -- e.g. 50 bps = 0.50% when via External SDK
    gst_rate_bps        INT NOT NULL DEFAULT 1800,  -- 18.00% GST on platform service fees
    effective_from      TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp(),
    effective_to        TIMESTAMPTZ,
    CONSTRAINT chk_statutory_split_100pct CHECK (
        split_auctioneer_bps + split_hms_or_market_bps +
        split_fisher_bonus_or_wf_bps + split_coop_or_lsgi_bps = 10000
    ),
    UNIQUE (tenant_id, version_no)
);

-- 2. VESSELS, FISHERMEN & SECTION 5 LICENSED AUCTIONEERS
CREATE TABLE harbour.vessels (
```

```
    vessel_id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id           UUID NOT NULL REFERENCES harbour.tenants(tenant_id),
    realcraft_reg_no    VARCHAR(64) NOT NULL,
    vessel_name         VARCHAR(128) NOT NULL,
    owner_user_id       UUID NOT NULL,
    nfdp_beneficiary_id VARCHAR(64),
    owner_razorpay_acct_id VARCHAR(64),
    -- Section 4(8) Kerala Act 16 of 2021: Tied Co-op Society Loan Encumbrance
    has_active_society_loan BOOLEAN NOT NULL DEFAULT FALSE,
    loan_coop_society_id UUID,
    UNIQUE (tenant_id, realcraft_reg_no)
);

CREATE TABLE harbour.auctioneer_licences (
    licence_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id           UUID NOT NULL REFERENCES harbour.tenants(tenant_id),
    auctioneer_user_id  UUID NOT NULL,
    licence_number      VARCHAR(64) NOT NULL,
    authorised_officer_ref VARCHAR(128) NOT NULL,
    deputed_by_society_id UUID, -- Required if auctioning for Section 4(8) loan-encumbered boats
    razorpay_account_id VARCHAR(64) NOT NULL,
    valid_from          DATE NOT NULL,
    valid_until         DATE NOT NULL, -- 3-year validity per Section 5(6)
    status              VARCHAR(24) NOT NULL CHECK (status IN ('ACTIVE','SUSPENDED','EXPIRED')),
    UNIQUE (tenant_id, licence_number)
);

-- 3. CATCH LOTS & WEIGHING RECORDS (Offline-Sync Ready via UUIDv7 + HLC)
CREATE TABLE catch_lot.fish_lots (
    lot_id              UUID PRIMARY KEY, -- Client-generated UUIDv7 for offline idempotency
    tenant_id           UUID NOT NULL REFERENCES harbour.tenants(tenant_id),
    vessel_id           UUID NOT NULL REFERENCES harbour.vessels(vessel_id),
    species_code        VARCHAR(32) NOT NULL,
    quality_grade       VARCHAR(24) NOT NULL CHECK (quality_grade IN ('A_EXPORT','A_DOMESTIC','B_PROCESS')),
    sale_unit_type      VARCHAR(16) NOT NULL CHECK (sale_unit_type IN ('WEIGHT_KG','BOX_COUNT','BASKET_COUNT','PIECE_COUNT')),
    gross_weight_grams  BIGINT NOT NULL CHECK (gross_weight_grams > 0),
    tare_weight_grams   BIGINT NOT NULL DEFAULT 0,
    net_weight_grams    BIGINT GENERATED ALWAYS AS (gross_weight_grams - tare_weight_grams) STORED,
```

```
    box_count               INT NOT NULL DEFAULT 1
    basic_floor_price_paise BIGINT NOT NULL  -- Snapshot of HMS Section 19 floor price at
weighing
    scale_ble_mac_address   VARCHAR 32
    is_manual_weight_entry  BOOLEAN NOT NULL DEFAULT FALSE
    qr_signature_hmac       VARCHAR 128  NOT NULL
    lot_status              VARCHAR 24  NOT NULL DEFAULT 'WEIGHED_READY'
                            CHECK (lot_status IN  'WEIGHED_READY'  'IN_AUCTION'  'SOLD_LOCKED'
'SETTLED'  'DISPATCHED'  'CANCELLED'))
    hlc_timestamp           VARCHAR 64  NOT NULL  -- Hybrid Logical Clock for offline sync ordering
    created_at              TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
)

-- 4. AUCTIONS, BIDS & CONFIRMED SALES (Single-Sale Guarantee BR-005)
CREATE TABLE auction.sales (
    sale_id                 UUID PRIMARY KEY DEFAULT gen_random_uuid()
    tenant_id               UUID NOT NULL REFERENCES harbour.tenants(tenant_id)
    lot_id                  UUID NOT NULL REFERENCES catch_lot.fish_lots(lot_id),
    auctioneer_licence_id   UUID NOT NULL REFERENCES
harbour.auctioneer_licences(licence_id),
    buyer_user_id           UUID NOT NULL
    policy_id               UUID NOT NULL REFERENCES harbour.commission_policies(policy_id),
    auction_mode            VARCHAR 32  NOT NULL
    source_channel          VARCHAR 24  NOT NULL CHECK (source_channel IN
'HARBOUR_FLOOR'  'REMOTE_APP'  'EXTERNAL_SDK'))
    hammer_price_paise      BIGINT NOT NULL CHECK (hammer_price_paise > 0)
    buyer_fee_paise         BIGINT NOT NULL DEFAULT 0
    gst_on_fee_paise        BIGINT NOT NULL DEFAULT 0
    buyer_total_payable_paise BIGINT GENERATED ALWAYS AS (hammer_price_paise +
buyer_fee_paise + gst_on_fee_paise) STORED
    idempotency_key         VARCHAR 128  UNIQUE NOT NULL
    confirmed_at            TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
    -- BR-005: A lot can NEVER be sold more than once
    CONSTRAINT uq_single_sale_per_lot UNIQUE (tenant_id, lot_id)
)

-- 5. FOOLPROOF DOUBLE-ENTRY LEDGER WITH CRYPTOGRAPHIC SHA-256 HASH CHAIN
CREATE TABLE ledger.accounts (
    account_id              UUID PRIMARY KEY DEFAULT gen_random_uuid()
    tenant_id               UUID NOT NULL REFERENCES harbour.tenants(tenant_id)
    owner_entity_type       VARCHAR 32  NOT NULL CHECK (owner_entity_type IN
                        'BUYER_ESCROW'  'BUYER_RECEIVABLE'  'FISHERMAN_PAYABLE'
                        'FISHERMAN_BONUS_ESCROW'  'AUCTIONEER_PAYABLE'  'HMS_PAYABLE'
```

```
                    'COOP_SOCIETY_PAYABLE', 'WELFARE_BOARD_PAYABLE', 'LSGI_PAYABLE',
                    'HARBOUROS_REVENUE', 'GST_TAX_PAYABLE'
                ),
    owner_entity_id     UUID        NOT NULL,
    currency_code       CHAR(3)     NOT NULL DEFAULT 'INR',
    UNIQUE (tenant_id, owner_entity_type, owner_entity_id)
);

CREATE TABLE ledger.journals (
    journal_id          UUID        PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id           UUID        NOT NULL REFERENCES harbourtenants(tenant_id),
    reference_type      VARCHAR(32) NOT NULL CHECK (reference_type IN (
                    'SALE_SETTLEMENT', 'BUYER_DEPOSIT', 'BANK_PAYOUT', 'AUTHORIZED_REVERSAL'
                )),
    reference_id        UUID        NOT NULL,  -- e.g. sale_id
    reverses_journal_id UUID        REFERENCES ledger.journals(journal_id),
    authorized_by_user_1 UUID       NOT NULL,
    authorized_by_user_2 UUID,      -- Required for reversals (Dual-control BR-008)
    reason_note         TEXT,
    created_at          TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
);

CREATE TABLE ledger.ledger_entries (
    entry_seq           BIGSERIAL   PRIMARY KEY,
    entry_id            UUID        UNIQUE NOT NULL DEFAULT gen_random_uuid(),
    tenant_id           UUID        NOT NULL REFERENCES harbourtenants(tenant_id),
    journal_id          UUID        NOT NULL REFERENCES ledger.journals(journal_id),
    account_id          UUID        NOT NULL REFERENCES ledger.accounts(account_id),
    direction           CHAR(1)     NOT NULL CHECK (direction IN ('D', 'C')),  -- D = Debit, C = Credit
    amount_paise        BIGINT      NOT NULL CHECK (amount_paise > 0),
    created_at          TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp(),
    prev_row_hash       CHAR(64)    NOT NULL,
    row_hash            CHAR(64)    NOT NULL
);

-- TRIGGER 1: Prevent ANY Update or Delete on Ledger Entries or Journals (Immutable Audit
BR-007)
CREATE OR REPLACE FUNCTION ledger.fn_prevent_ledger_tampering()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
BEGIN
    RAISE EXCEPTION 'TAMPER_PROTECTION_VIOLATION: Table %.% is strictly append-only. Use a
compensating reversal journal.'
        TG_TABLE_SCHEMA, TG_TABLE_NAME;
```

```
END
$$;

CREATE TRIGGER trg_immutable_ledger_entries
BEFORE UPDATE OR DELETE ON ledger.ledger_entries
FOR EACH ROW EXECUTE FUNCTION ledger.fn_prevent_ledger_tampering();

CREATE TRIGGER trg_immutable_journals
BEFORE UPDATE OR DELETE ON ledger.journals
FOR EACH ROW EXECUTE FUNCTION ledger.fn_prevent_ledger_tampering();

-- TRIGGER 2: Compute Cryptographic SHA-256 Hash Chain per Tenant on Insert
CREATE OR REPLACE FUNCTION ledger.fn_compute_ledger_hash_chain()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
DECLARE
    v_last_hash CHAR(64);
    v_payload TEXT;
BEGIN
    -- Lock tenant's latest ledger sequence to serialize hash chaining cleanly
    SELECT row_hash INTO v_last_hash
    FROM ledger.ledger_entries
    WHERE tenant_id = NEW.tenant_id
    ORDER BY entry_seq DESC
    LIMIT 1
    FOR UPDATE;

    IF v_last_hash IS NULL THEN
        v_last_hash = repeat('0', 64); -- Genesis hash for tenant
    END IF;

    NEW.prev_row_hash = v_last_hash;
    v_payload = concat_ws('|',
        NEW.tenant_id::TEXT,
        NEW.prev_row_hash,
        NEW.journal_id::TEXT,
        NEW.account_id::TEXT,
        NEW.direction,
        NEW.amount_paise::TEXT,
        extract(epoch from NEW.created_at)::TEXT
    );
    NEW.row_hash = encode(digest(v_payload, 'sha256'), 'hex');
    RETURN NEW;
END
```

```
$$;

CREATE TRIGGER trg_hash_chain_ledger_insert
BEFORE INSERT ON ledger.ledger_entries
FOR EACH ROW EXECUTE FUNCTION ledger.fn_compute_ledger_hash_chain();

-- TRIGGER 3: Deferred Constraint Trigger Verifying Double-Entry Balance (SUM(D) == SUM(C))
CREATE OR REPLACE FUNCTION ledger.fn_verify_journal_balanced()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
DECLARE
    v_debits BIGINT;
    v_credits BIGINT;
BEGIN
    SELECT
        COALESCE(SUM(CASE WHEN direction = 'D' THEN amount_paise ELSE 0 END), 0),
        COALESCE(SUM(CASE WHEN direction = 'C' THEN amount_paise ELSE 0 END), 0)
    INTO v_debits, v_credits
    FROM ledger.ledger_entries
    WHERE journal_id = NEW.journal_id;

    IF v_debits <> v_credits OR v_debits = 0 THEN
        RAISE EXCEPTION 'UNBALANCED_JOURNAL_VIOLATION: Journal % has Debits=% paise, Credits=% paise',
            NEW.journal_id, v_debits, v_credits;
    END IF;
    RETURN NULL;
END;
$$;

CREATE CONSTRAINT TRIGGER trg_enforce_double_entry_balance
AFTER INSERT ON ledger.ledger_entries
DEFERRABLE INITIALLY DEFERRED
FOR EACH ROW EXECUTE FUNCTION ledger.fn_verify_journal_balanced();

-- 6. ROW-LEVEL SECURITY (RLS) ENFORCEMENT FOR MULTI-TENANT ISOLATION (NFR-023)
ALTER TABLE catch_lot.fish_lots ENABLE ROW LEVEL SECURITY;
ALTER TABLE auction.sales ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger.ledger_entries ENABLE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation_policy_lots ON catch_lot.fish_lots
    USING (
        current_setting('app.current_user_role', true) = 'SUPER_ADMIN'
        OR tenant_id = current_setting('app.current_tenant_id', true)::UUID
```

```
);

CREATE POLICY tenant_isolation_policy_sales ON auction.sales
  USING (
    current_setting('app.current_user_role', true) = 'SUPER_ADMIN'
    OR tenant_id = current_setting('app.current_tenant_id', true)::UUID
);

CREATE POLICY tenant_isolation_policy_ledger ON ledger.ledger_entries
  USING (
    current_setting('app.current_user_role', true) = 'SUPER_ADMIN'
    OR tenant_id = current_setting('app.current_tenant_id', true)::UUID
);
```

# 4. External SDK, API & Multi-Party Split Integration Code

## 4.1 Embeddable External SDK (@harbouros/auction-sdk)

When an external harbour portal or seafood exchange wants to host auctions on its own site while using HarbourOS's real-time bidding clock, buyer credit checks, and split-settlement rails, it embeds @harbouros/auction-sdk:

```
TypeScript
// External Partner Website Integration Example
import { HarbourOSClient, mountAuctionClockWidget } from '@harbouros/auction-sdk';

const harbourClient = new HarbourOSClient({
  publishableKey: 'pk_live_harbouros_kerala_kochi_99281',
  tenantId: '7c9e6679-7425-40de-944b-e07fc1f90ae7',
  buyerSessionToken: 'eyJhbGciOiJFZERTQSI...', // Issued via server-to-server POST /v1/sdk/buyer-sessions
  locale: 'ml-IN', // Supports 'ml-IN', 'en-IN', 'ja-JP', 'es-ES'
});

const clockWidget = mountAuctionClockWidget('#external-harbour-auction-container', {
  client: harbourClient,
  harbourRoomId: 'kochi-shed-01',
  mode: 'HYBRID_CLOCK_AND_SHOUT',
  onLotWon: (saleReceipt) =>
```

```
  // Automatically triggers HarbourOS Split Settlement Checkout (with our SDK fee cut built in)
  harbourClient.openSplitSettlementCheckout(saleReceipt.saleId);
}
}
```

## 4.2 Deterministic Multi-Party Split Calculator (packages/domain-money)

Inside svc-ledger-settlement, once the ledger_entries transaction commits, the settlement service computes the exact paise transfers for **Razorpay Route** (or Cashfree Easy Split / Stripe Connect)[6]. Note that exact Razorpay REST payload parameter names should be bound against the live Razorpay Route SDK version during integration, while the underlying transfer math is strictly verified below[6]:

TypeScript

```
// packages/domain-money/src/kerala-act16-split.calculator.ts
export interface KeralaHarbourSplitInput {
  saleId: string;
  hammerPricePaise: number;       // e.g., 10_000_000 (₹1,00,000.00)
  sellerCommBps: number;          // e.g., 500 (5.00% statutory cap under Act 16 Sec 4(4))
  harbourosCutFromHmsBps: number; // e.g., 2000 (20% of HMS share = 20 bps of Hammer)
  harbourosCutFromCoopBps: number; // e.g., 2000 (20% of Co-op share = 20 bps of Hammer)
  buyerClearingFeeBps: number;    // e.g., 100 (1.00% Buyer Clearing / SDK Fee)
  gstRateBps: number;             // e.g., 1800 (18.00% GST on platform fee)
}

export function calculateKeralaHarbourSplit(input: KeralaHarbourSplitInput) {
  const totalSellerCommPaise = Math.floor((input.hammerPricePaise * input.sellerCommBps) / 10000);
  const fishermanInstantNetPaise = input.hammerPricePaise - totalSellerCommPaise;

  // Section 4(5)(a) 4-Way Apportionment of the 5% Commission:
  const auctioneerNetPaise = Math.floor((totalSellerCommPaise * 2000) / 10000); // 20%
  const hmsGrossPaise = Math.floor((totalSellerCommPaise * 2000) / 10000);       // 20%
  const bonusEscrowPaise = Math.floor((totalSellerCommPaise * 4000) / 10000);    // 40%
  const coopGrossPaise =
    totalSellerCommPaise - (auctioneerNetPaise + hmsGrossPaise + bonusEscrowPaise); // 20%

  // HarbourOS Variable B2B Cut (Deducted from HMS and Co-op Society shares)
```

```
  const platformFromHmsPaise = Math.floor((hmsGrossPaise * input.harbourosCutFromHmsBps) / 10000);
  const platformFromCoopPaise = Math.floor((coopGrossPaise * input.harbourosCutFromCoopBps) / 10000);

  const hmsNetPaise = hmsGrossPaise - platformFromHmsPaise;
  const coopNetPaise = coopGrossPaise - platformFromCoopPaise;

  // Buyer-Side Clearing / External SDK Fee
  const buyerFeePaise = Math.floor((input.hammerPricePaise * input.buyerClearingFeeBps) / 10000);
  const gstOnBuyerFeePaise = Math.floor((buyerFeePaise * input.gstRateBps) / 10000);
  const buyerTotalPayablePaise = input.hammerPricePaise + buyerFeePaise + gstOnBuyerFeePaise;

  const harbourosTotalRevenuePaise = platformFromHmsPaise + platformFromCoopPaise + buyerFeePaise;

  return {
    buyerTotalPayablePaise,      // 10,118,000 paise (₹1,01,180.00)
    fishermanInstantNetPaise,    //  9,500,000 paise (₹95,000.00)
    bonusEscrowPaise,            //    200,000 paise (₹2,000.00)
    auctioneerNetPaise,          //    100,000 paise (₹1,000.00)
    hmsNetPaise,                 //     80,000 paise (₹800.00)
    coopNetPaise,                //     80,000 paise (₹800.00)
    harbourosTotalRevenuePaise,  //    140,000 paise (₹1,400.00)
    gstOnBuyerFeePaise,          //     18,000 paise (₹180.00)
  };
}
```

# 5. Phased Build Plan & Master AI Engineering Prompt

## 5.1 Six-Month Engineering & Pilot Execution Roadmap

1. **Month 1 — Field Discovery, Legal Validation & Monorepo Scaffolding**: Visit a target Kerala fishing harbour to observe at least 10 live morning auction transactions and interview HMS officials, licensed auctioneers, and fishermen[1]. Initialize the Nx Monorepo, docker-compose.dev.yml (PostgreSQL 16, PgBouncer, Redis 7, NATS JetStream), and the packages/domain-money and packages/db-rls-driver libraries.
2. **Month 2 — Database DDL, Hash-Chain Ledger & Core Services (svc-identity, svc-harbour)**: Deploy the PostgreSQL 16 schemas, RLS policies, and SHA-256 append-only ledger triggers[1]. Build passwordless OTP/WebAuthn login, Section 5 Auctioneer Licence verification, Section 19 Basic Floor Price tables, and Section 4(8)

Co-op Loan vessel rules[5].

3. **Month 3 — Offline-First Catch & Weighing Suite (svc-catch-lot + Flutter App)**: Build the bilingual (ml-IN / en-IN) Flutter mobile app with local SQLite (SQLCipher) offline storage, BLE scale weight locking, and Bluetooth thermal QR label printing[1].
4. **Month 4 — Real-Time Hybrid Auction Engine (svc-auction-engine in Go) & Split Settlement (svc-ledger-settlement)**: Build the Golang WebSocket/gRPC auction engine supporting Physical Shout, Hybrid Live, and Dutch Descending Clock modes, wired via NATS JetStream to the double-entry ledger and Razorpay Route split settlement pipeline[3].
5. **Month 5 — Super Admin Control Plane, @harbouros/auction-sdk & Hardening**: Build the Next.js Super Admin and Harbour Admin dashboards, package the embeddable external SDK and webhook gateway (svc-sdk-gateway), and run chaos testing on offline sync conflict resolution and ledger immutability[1].
6. **Month 6 — On-Site Pilot Rollout at One Kerala Harbour / Landing Centre**: Install dockside BLE weighing bridges and thermal printers, train auctioneers and weighing operators, shadow 100+ live lots, and validate daily settlement reconciliation with the Harbour Management Society[1].

### 5.2 Copy-Pasteable Master AI Engineering Prompt

Copy and paste the following **Master System Prompt** into Cursor, Claude Code, Windsurf, or GitHub Copilot Workspace to scaffold and build the entire HarbourOS codebase systematically:

# SYSTEM PROMPT: HARBOUROS PRINCIPAL DISTRIBUTED SYSTEMS & FINTECH ARCHITECT

You are the Principal Architect and Lead Systems Engineer for **HarbourOS**, a multi-tenant, cloud-edge Fisheries Harbour Operations, Real-Time Hybrid Auction, and Tamper-Proof Financial Clearing SaaS platform and embeddable SDK ecosystem.

## 1. ARCHITECTURAL INVARIANTS (NEVER VIOLATE)

1. **Monorepo & Clean Hexagonal Boundaries**: All code lives in an Nx Monorepo (apps/* for microservices/frontends, packages/* for shared libraries). Services communicate synchronously ONLY via gRPC (packages/proto-contracts) and asynchronously via NATS JetStream using the Transactional Outbox pattern. Cross-service database joins or direct cross-app TypeScript imports are strictly forbidden.
2. **Multi-Tenant Isolation (PostgreSQL 16 RLS + PgBouncer)**: Every operational table contains a tenant_id UUID NOT NULL column enforced by PostgreSQL Row-Level Security (RLS). Every database query MUST execute inside TenantAwareDbPool.withTenantTransaction(), which sets set_config('app.current_tenant_id', tenantId, true) locally per transaction so PgBouncer

(pool_mode = transaction) never leaks tenant context.

3. **Integer Monetary Math & Double-Entry Conservation**: Never use FLOAT, DOUBLE, or JavaScript Number floats for money or weights. Store all currency in BIGINT smallest units (paise for INR, cents for USD/EUR) and weights in BIGINT grams. All commission splits use integer Basis Points (10,000 bps = 100.00%) with deterministic Largest-Remainder allocation so SUM(allocated_shares) === total_amount.
4. **Cryptographic Append-Only Ledger**: The ledger.ledger_entries and ledger.journals tables are strictly append-only (BEFORE UPDATE OR DELETE raises an exception). Every journal must satisfy SUM(Debit) === SUM(Credit) via deferred constraint trigger, and every entry links to the tenant's previous entry via a SHA-256 hash chain (prev_row_hash -> row_hash).
5. **Kerala Act 16 of 2021 Statutory Compliance & Variable Monetization**:
   - Enforce Section 5 Auctioneer Licences (block expired licences or wrong harbour locations).
   - Enforce Section 4(8) Co-op Loan Encumbrances (vessels with has_active_society_loan = true can only be auctioned by an auctioneer with matching deputed_by_society_id).
   - Enforce Section 19 HMS Basic Floor Prices (bid_paise >= basic_floor_price_paise).
   - Enforce Section 4(4) & 4(5) Seller Commission Apportionment (max 500 bps split 20% Auctioneer, 20% HMS/FLCMS, 40% Fisherman Annual Bonus Escrow, 20% Co-op Society at harbours; or 20/20/30/30 at fish markets).
   - Extract HarbourOS's variable revenue cut via two configurable levers per tenant (harbour.commission_policies): (a) B2B Institutional Tech-Partner Share deducted from the HMS and Society shares (harbouros_cut_from_hms_bps, harbouros_cut_from_coop_bps), and (b) Buyer-Side Clearing / External SDK Fee (buyer_clearing_fee_bps, sdk_external_api_fee_bps) added on top of the hammer price, settled automatically via Razorpay Route / Stripe Connect.
6. **Offline-First Edge Ergonomics**: All field mobile workflows (Catch Weighing, Lot QR Printing, Floor Shout Recording) must operate 100% offline using local SQLite (SQLCipher), UUIDv7 primary keys, Hybrid Logical Clocks (hlc_timestamp), and idempotent sync queues (idempotency_key), with bilingual Malayalam (ml-IN) and English (en-IN) UI and voice-readout support.

# 2. EXECUTION TASK SEQUENCE

When asked to build a module, generate production-grade, fully typed, zero-placeholder code in this exact order:

1. Shared Protobuf / TypeScript domain contracts (packages/proto-contracts, packages/domain-money).
2. PostgreSQL 16 SQL migrations, RLS policies, and triggers (infra/postgres-migrations).
3. Backend microservice implementation with unit and integration tests (apps/svc-*).
4. Frontend / Mobile / SDK client implementation (apps/web-*, apps/mobile-harbour-edge, packages/auction-sdk-js).

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